

```
In [40]: import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
import pandas as pd
import seaborn as sns
import numpy as np
from collections import defaultdict
from scipy.cluster.hierarchy import dendrogram, linkage
from scipy.spatial.distance import squareform
from Levenshtein import distance as levenshtein_distance
import networkx as nx
import math
from rich import print as rprint # For better terminal formatting
```

```
In [2]: %load_ext sql

%sql postgresql://postgres:EMvAYOrD%23BYU8y@localhost/political_finance_data
%config SqlMagic.style = 'PLAIN_COLUMNS'
%config SqlMagic.autopandas = True
```

 Shadow Network Intelligence Report

Dark Money & Out-of-State Influence in Andy Barr's Campaign Financing

Executive Summary Summary of key findings:

- Use of WinRed and Friends of Andy Barr to route undisclosed donations.
- Many large donations each over \$10,000, not even bothering to hide being over the limit.
- Those making large donations also seem to donate in networks clustered by family and employer.
- Significant financial activity from Georgia-based committees.
- Clear callout: "Who is funding Andy Barr, and why does it matter?"

Andy Barr has three active committees named after him

```
In [277... %%sql
SELECT COMMITTEE_ID, NAME, STATE, PARTY, CANDIDATE_IDS, cycles
FROM COMMITTEES
WHERE name ILIKE '%Andy Barr%'

* postgresql://postgres:***@localhost/political_finance_data
4 rows affected.
```

Out[277...

	committee_id	name	state	party	candidate_ids	cycles
0	C00467571	ANDY BARR FOR SENATE, INC.	KY	REP	[H0KY06104, S6KY00286]	[2010, 2012, 2014, 2016, 2018, 2020, 2022, 202...
1	C00524728	ANDY BARR VICTORY 2012	KY	None	[H0KY06104]	[2012, 2014]
2	C00618777	ANDY BARR VICTORY COMMITTEE	GA	None	[]	[2016, 2018, 2020, 2022, 2024, 2026]
3	C00633792	FRIENDS OF ANDY BARR COMMITTEE	GA	None	[]	[2018, 2020, 2022, 2024, 2026]

1. Andy Barr’s Fundraising Overview (election cycle 2026)

Purpose: Contextualize total fundraising by Month, year and source type.

 Chart 1: Fundraising Over Time

- Title: “Andy Barr’s Total Fundraising by Month”

In [169...

```
contributions = %sql SELECT entity_type_desc AS type \
, EXTRACT(MONTH FROM contribution_receipt_date) AS month \
, EXTRACT(YEAR FROM contribution_receipt_date) AS year \
, SUM(contribution_receipt_amount) AS total_raised \
FROM schedule_a_contributions \
WHERE committee_id in ('C00467571', 'C00618777', 'C00633792') \
GROUP BY entity_type_desc, year, month \
ORDER BY year, month;

* postgresql://postgres:***@localhost/political_finance_data
13 rows affected.
```

In [170...

```
contributions
```

Out[170...

	type	month	year	total_raised
0	POLITICAL ACTION COMMITTEE	12	2024	1500.00
1	INDIVIDUAL	1	2025	136275.10
2	ORGANIZATION	1	2025	41507.64
3	POLITICAL ACTION COMMITTEE	1	2025	27574.25
4	INDIVIDUAL	2	2025	470460.83
5	ORGANIZATION	2	2025	22039.26
6	OTHER COMMITTEE	2	2025	36560.76
7	POLITICAL ACTION COMMITTEE	2	2025	129487.62
8	CAMPAIGN COMMITTEE	3	2025	4000.00
9	INDIVIDUAL	3	2025	1324108.24
10	ORGANIZATION	3	2025	84106.29
11	OTHER COMMITTEE	3	2025	784915.78
12	POLITICAL ACTION COMMITTEE	3	2025	851960.51

In [171...

```

# Apply a clean theme
sns.set_theme(style="whitegrid") # alternatives: "darkgrid", "white", "ticks"

plt.rcParams['axes.titlesize'] = 16
plt.rcParams['axes.labelsize'] = 12
plt.rcParams['legend.fontsize'] = 10
plt.rcParams['legend.title_fontsize'] = 11
plt.rcParams['xtick.labelsize'] = 10
plt.rcParams['ytick.labelsize'] = 10

# Increase figure size globally
plt.rcParams['figure.figsize'] = (12, 6)

# Convert the result to a pandas DataFrame -- this is done automatically because I
contributions_df = contributions

# contributions_df should have: entity_type_desc, month, year, total_raised
contributions_df['date'] = pd.to_datetime(
    contributions_df['year'].astype(int).astype(str) + '-' +
    contributions_df['month'].astype(int).astype(str).str.zfill(2) + '-01'
)

# Ensure total_raised is numeric
contributions_df['total_raised'] = pd.to_numeric(contributions_df['total_raised'],

# Pivot so each entity_type_desc is a line
pivot = contributions_df.pivot_table(
    index='date',
    columns='type',
    values='total_raised',

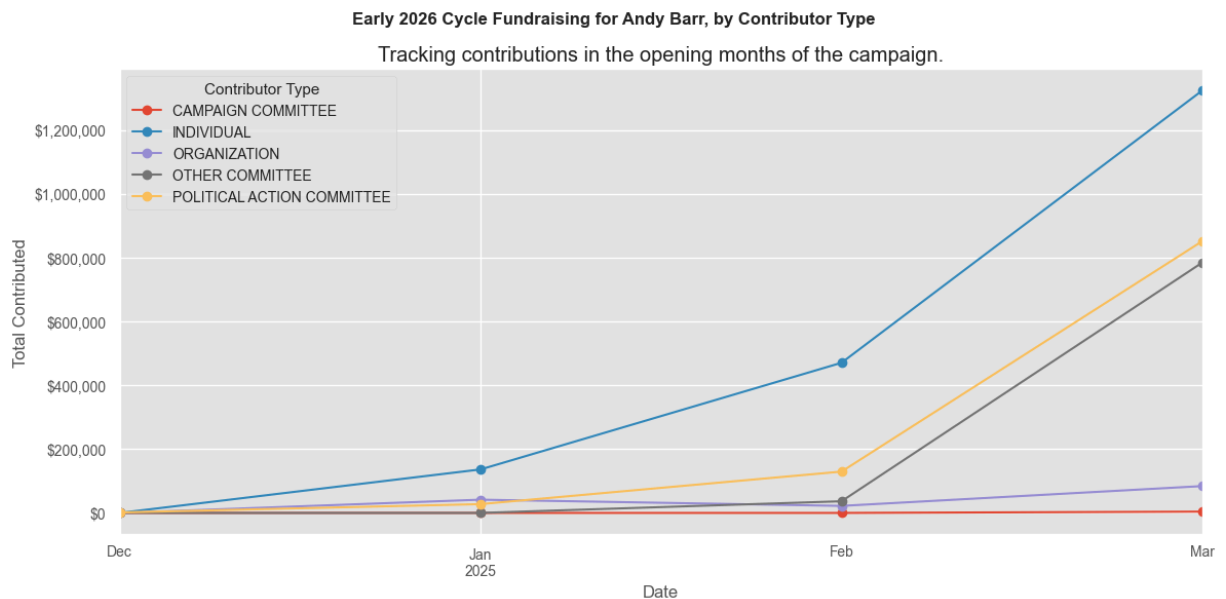
```

```

aggfunc='sum'
).fillna(0)

plt.figure(figsize=(12, 6))
plt.style.use("ggplot")
pivot.plot(ax=plt.gca(), marker='o')
plt.suptitle("Early 2026 Cycle Fundraising for Andy Barr, by Contributor Type", fontweight="bold")
plt.title("Tracking contributions in the opening months of the campaign.")
plt.xlabel("Date")
plt.gca().yaxis.set_major_formatter(ticker.FuncFormatter(lambda x, _: f'${x:,.0f}'))
plt.ylabel("Total Contributed")
plt.legend(title="Contributor Type")
plt.grid(True)
plt.tight_layout()
plt.show()

```



Top 10 Contributors to Andy Barr's committees

In [264...

```

%%sql
SELECT contributor_name, SUM(contribution_receipt_amount) AS total
FROM schedule_a_contributions WHERE committee_id in ('C00467571', 'C00618777', 'C00
-- AND contributor_name ILIKE '%WinRed%'
GROUP BY contributor_name
ORDER BY total DESC
LIMIT 10;

```

* postgresql://postgres:***@localhost/political_finance_data
10 rows affected.

Out [264...

	contributor_name	total
0	FRIENDS OF ANDY BARR	528791.68
1	WINRED	491150.50
2	FORCHT BANK	54386.61
3	CALLER, STEVE	41000.00
4	MOCK, ERIC	31000.00
5	BROWN, JACK	31000.00
6	REDDY, SHASHI	31000.00
7	JAIN, ANKUR	31000.00
8	HAAG, TIM	31000.00
9	WILLIAMS, DAVID	31000.00

WinRed and Friends of Andy Barr are the top two contributors to Andy Barr's campaign

Together these two donors account for 26% of Andy Barr's receipts. And they are completely opaque - obscuring all information about the donors.

contributor	entity type	contributions	amount
FRIENDS OF ANDY BARR	OTHER COMMITTEE	3	\$528,791.68
WINRED	OTHER COMMITTEE	45	\$271,128.12
WINRED	POLITICAL ACTION COMMITTEE	1134	\$220,022.38

2. WinRed and Friends of Andy Barr Contributions & Lack of Donor Disclosure

Purpose: Show how much was funneled through opaque organizations, and the difficulty of tracing donor origins.

 Chart 2: Contributions via WinRed

- Title: "Funds Raised via WinRed Platform"
- Chart Type: Bar chart

In [263...

```
%%sql
WITH total_fundraising AS (
  SELECT SUM(contribution_receipt_amount) AS grand_total
  FROM schedule_a_contributions
```

```

WHERE committee_id IN ('C00467571', 'C00618777', 'C00633792')
)
SELECT
  contributor_name,
  TO_CHAR(SUM(contribution_receipt_amount), '$999,999,999') AS total,
  COUNT(*) AS count,
  TO_CHAR(SUM(contribution_receipt_amount)/COUNT(*), '$999,999,999') AS avg,
  ROUND(100.0 * SUM(contribution_receipt_amount) / total_fundraising.grand_total, 2
FROM schedule_a_contributions, total_fundraising
WHERE committee_id IN ('C00467571', 'C00618777', 'C00633792')
  AND contributor_name ILIKE '%WinRed%' or contributor_name ILIKE '%Friends of Andy
GROUP BY contributor_name, total_fundraising.grand_total
ORDER BY total DESC
LIMIT 50;

```

* postgresql://postgres:***@localhost/political_finance_data
2 rows affected.

Out[263...

	contributor_name	total	count	avg	pct_of_total
0	FRIENDS OF ANDY BARR	\$ 528,792	3	\$ 176,264	13.51
1	WINRED	\$ 491,151	1179	\$ 417	12.55

Bar chart comparing missing data rates (employer & occupation) between WinRed and all other donors.



Step-by-step Plan:

We'll:

- 1. Create three grouped queries: one for WinRed donors, one for Friends of Andy Barr donors, one for everyone else.
- 2. Calculate % missing employer and % missing occupation in each group.
- 3. Visualize with a side-by-side bar chart.



SQL Logic Summary

We'll define:

- "Missing" = NULL, blank, "N/A", "NONE", "NOT EMPLOYED", "INFORMATION REQUESTED", etc.
- Each group's total donors and how many are missing occupation/employer.

In [267...

```

# Define your helper list of missing values
missing_values = {"", "N/A", "NONE", "NOT EMPLOYED", "INFORMATION REQUESTED", "SELF

# Query 1 - WinRed donors

```

```

winred_stats = %sql SELECT \
    COUNT(*) FILTER ( \
        WHERE TRIM(UPPER(contributor_employer)) NOT IN ('', 'N/A', 'NONE', 'NOT EMP
    )::float / COUNT(*) AS pct_employer_provided, \
    COUNT(*) FILTER ( \
        WHERE TRIM(UPPER(contributor_occupation)) NOT IN ('', 'N/A', 'NONE', 'NOT E
    )::float / COUNT(*) AS pct_occupation_provided \
FROM schedule_a_contributions \
WHERE committee_id IN ('C00467571', 'C00618777', 'C00633792') \
    AND contributor_name ILIKE '%WinRed%';

# Query 2 - Friends of Andy Barr donors
friends_stats = %sql SELECT \
    COUNT(*) FILTER ( \
        WHERE TRIM(UPPER(contributor_employer)) NOT IN ('', 'N/A', 'NONE', 'NOT EMP
    )::float / COUNT(*) AS pct_employer_provided, \
    COUNT(*) FILTER ( \
        WHERE TRIM(UPPER(contributor_occupation)) NOT IN ('', 'N/A', 'NONE', 'NOT E
    )::float / COUNT(*) AS pct_occupation_provided \
FROM schedule_a_contributions \
WHERE committee_id IN ('C00467571', 'C00618777', 'C00633792') \
    AND contributor_name ILIKE '%Friends of Andy Barr%';

# Query 3 - All other donors
other_stats = %sql SELECT \
    COUNT(*) FILTER ( \
        WHERE TRIM(UPPER(contributor_employer)) NOT IN ('', 'N/A', 'NONE', 'NOT EMP
    )::float / COUNT(*) AS pct_employer_provided, \
    COUNT(*) FILTER ( \
        WHERE TRIM(UPPER(contributor_occupation)) NOT IN ('', 'N/A', 'NONE', 'NOT E
    )::float / COUNT(*) AS pct_occupation_provided \
FROM schedule_a_contributions \
WHERE committee_id IN ('C00467571', 'C00618777', 'C00633792') \
    AND contributor_name NOT ILIKE '%WinRed%' \
    AND contributor_name NOT ILIKE '%Friends of Andy Barr%';

```

```

* postgresql://postgres:***@localhost/political_finance_data
1 rows affected.
* postgresql://postgres:***@localhost/political_finance_data
1 rows affected.
* postgresql://postgres:***@localhost/political_finance_data
1 rows affected.

```

In [271...

```

import numpy as np
import matplotlib.pyplot as plt

# Assuming winred_stats, friends_stats, and other_stats are DataFrames with the nec
# Extract values as flat lists
winred_vals = winred_stats.iloc[0].tolist()
friends_vals = friends_stats.iloc[0].tolist()
other_vals = other_stats.iloc[0].tolist()

labels = ['Employer Provided', 'Occupation Provided']
x = np.arange(len(labels))
bar_width = 0.22

```

```

fig, ax = plt.subplots(figsize=(8, 5))

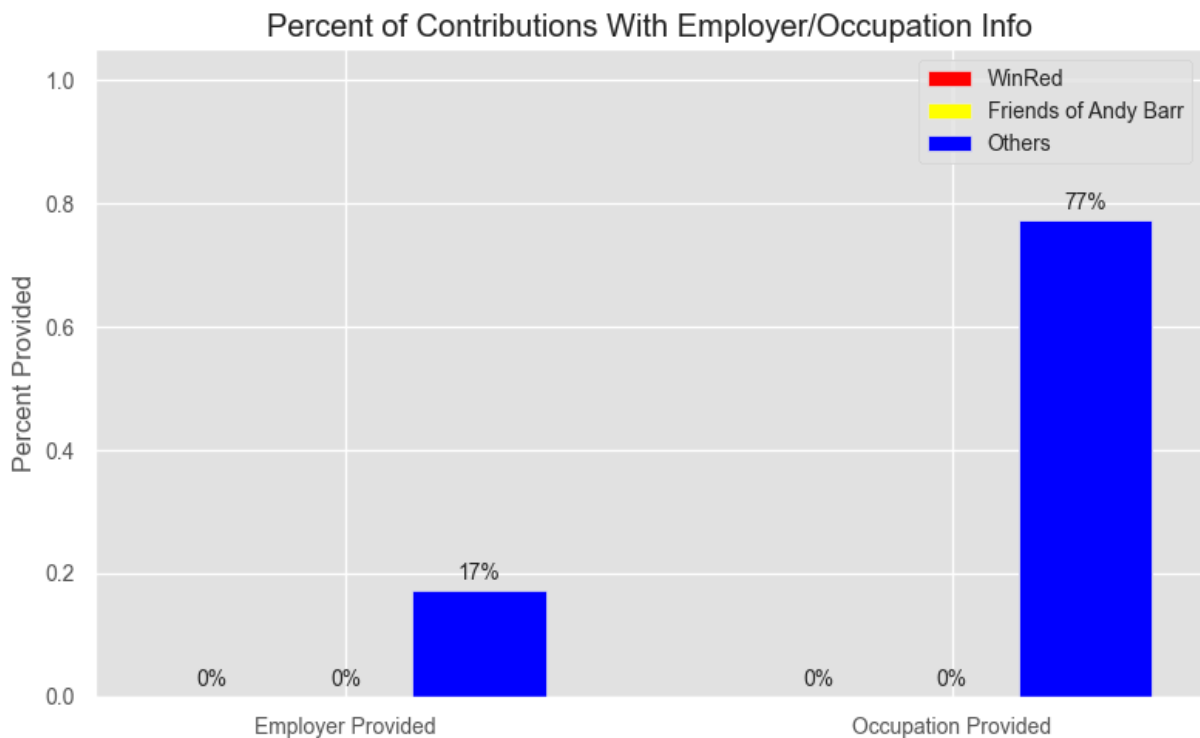
# Plotting the bars
bars1 = ax.bar(x - bar_width, winred_vals, bar_width, label='WinRed', color='red')
bars2 = ax.bar(x, friends_vals, bar_width, label='Friends of Andy Barr', color='yellow')
bars3 = ax.bar(x + bar_width, other_vals, bar_width, label='Others', color='blue')

# Add percentage labels above bars
for bars in [bars1, bars2, bars3]:
    for bar in bars:
        height = bar.get_height()
        ax.text(bar.get_x() + bar.get_width() / 2,
                height + 0.01,
                f'{height:.0%}',
                ha='center', va='bottom', fontsize=10)

# Setting labels and title
ax.set_ylabel('Percent Provided')
ax.set_title('Percent of Contributions With Employer/Occupation Info')
ax.set_xticks(x)
ax.set_xticklabels(labels)
ax.set_ylim(0, 1.05)
ax.legend()

plt.tight_layout()
plt.show()

```



Interpretation tip:

WinRed's zero values are not a data error — it's a reporting pattern, and that omission might reflect systemic evasion of disclosure norms.

 Text placeholder: Explain how WinRed aggregates contributions, often obscuring the original donor. Note any lack of itemization or incomplete records.

Note: WinRed-contributed donations consistently show no employer or occupation information. This is not due to missing data but appears to reflect a deliberate pattern of incomplete disclosure. By contrast, the majority of other donations provide this legally required information.

(Table) for transparency and storytelling.

In [256...

```
import pandas as pd

# Extract the values from the query results
winred_pct_employer = winred_stats.iloc[0, 0]
winred_pct_occupation = winred_stats.iloc[0, 1]

friends_pct_employer = friends_stats.iloc[0, 0]
friends_pct_occupation = friends_stats.iloc[0, 1]

other_pct_employer = other_stats.iloc[0, 0]
other_pct_occupation = other_stats.iloc[0, 1]

# Create the table data
data = {
    "Group": ["WinRed", "Friends of Andy Barr", "Other Donors"],
    "% Employer Provided": [f"{winred_pct_employer:.0%}", f"{friends_pct_employer:.0%}", f"{other_pct_employer:.0%}"],
    "% Occupation Provided": [f"{winred_pct_occupation:.0%}", f"{friends_pct_occupation:.0%}", f"{other_pct_occupation:.0%}"]
}

# Create and display the DataFrame
disclosure_df = pd.DataFrame(data)
display(disclosure_df)

# Optional footnote
print("\nNote: WinRed contributions consistently show no employer or occupation information. This appears to reflect a pattern of incomplete disclosure rather than a data error.")
```

	Group	% Employer Provided	% Occupation Provided
0	WinRed	0%	0%
1	Friends of Andy Barr	0%	0%
2	Other Donors	17%	77%

Note: WinRed contributions consistently show no employer or occupation information. This appears to reflect a pattern of incomplete disclosure rather than a data error.

Next, I just want to see how much they are donating compared to other types of entities

```
In [265... top_contributions = %sql \
SELECT \
    CASE \
        WHEN UPPER(contributor_name) ILIKE '%FRIENDS%' THEN 'Friends of Andy Barr'
        WHEN UPPER(contributor_name) ILIKE '%WINRED%' THEN 'WinRed' \
        ELSE entity_type_desc \
    END AS donor_group, \
    SUM(contribution_receipt_amount) AS total \
FROM schedule_a_contributions \
WHERE committee_id IN ('C00467571', 'C00618777', 'C00633792') \
GROUP BY donor_group \
ORDER BY total DESC;

top_contributions
```

* postgresql://postgres:***@localhost/political_finance_data
7 rows affected.

```
Out[265...

```

	donor_group	total
0	INDIVIDUAL	1930844.17
1	POLITICAL ACTION COMMITTEE	790500.00
2	Friends of Andy Barr	528791.68
3	WinRed	491150.50
4	ORGANIZATION	147653.19
5	OTHER COMMITTEE	21556.74
6	CAMPAIGN COMMITTEE	4000.00

Bar Chart comparing WinRed and Friends of Andy Barr Contributions to those by other entity types

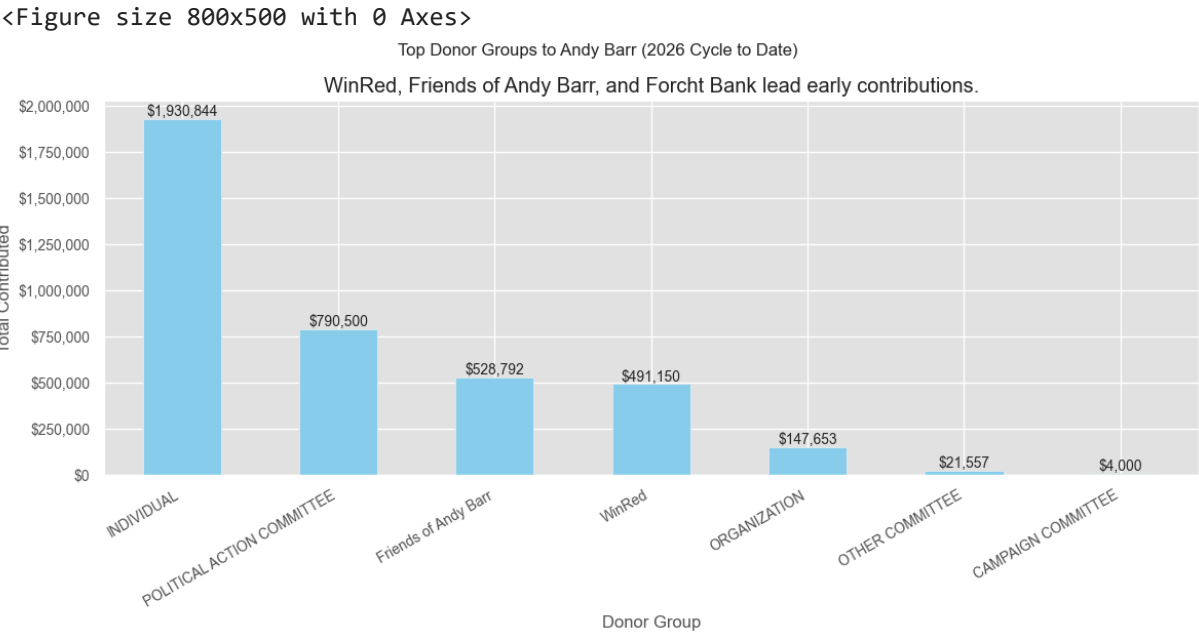
```
In [266... top_contributions_df = top_contributions

# Ensure 'total' is numeric
top_contributions_df['total'] = pd.to_numeric(top_contributions_df['total'], errors

# Sort for better visualization
top_contributions_df = top_contributions_df.sort_values('total', ascending=False)

# Plot
plt.figure(figsize=(8, 5))
top_contributions_df.plot(kind='bar', x='donor_group', y='total', legend=False, col
plt.gca().yaxis.set_major_formatter(ticker.FuncFormatter(lambda x, _: f'${x:,.0f}'))
for index, value in enumerate(top_contributions_df['total']):
    plt.text(index, value, f'${value:,.0f}', ha='center', va='bottom', fontsize=10)
plt.suptitle("Top Donor Groups to Andy Barr (2026 Cycle to Date)")
plt.title("WinRed, Friends of Andy Barr, and Forcht Bank lead early contributions.")
plt.ylabel("Total Contributed")
plt.xlabel("Donor Group")
```

```
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.show()
```



ADDITIONAL CHARTS showing networks of giving

Charting Coordinated Donor Patterns

To visualize potential dark money and coordinated giving, we’ll generate multiple **network charts**, each focused on a specific coordination signal. This avoids clutter and makes it easier to interpret complex patterns.

 Chart Options

Chart Type	Coordination Signal	Examples
 Last Name Clustering	Family or household-level bundling	Becks, Wilburns, Rowans
 Employer-Based Network	Shared workplace or firm-wide giving	Hedge funds, law firms, lobby shops
 LLC or Entity Bundling	Contributions from opaque corporate entities	Caller Properties LLC, Fortune Trade Center
 Shared Address or Zip	Geographic clustering or hosted PAC events	Donors from same building or city block

 Strategy

We'll start by creating **separate network charts** for each type of coordination. This helps keep the visuals clean, avoids information overload, and supports clear storytelling in our reports.

Later, we may explore combining them into a **composite interactive graph** using `Plotly` or `D3.js`, but for now we'll focus on individual patterns.

Last Name Clustering

```
In [242... # --- STEP 1: Query contributions and extract last names ---
family_contributions = %sql \
    SELECT \
        INITCAP(SPLIT_PART(contributor_name, ',', 1)) AS last_name, \
        contributor_name, \
        contributor_state, \
        SUM(contribution_receipt_amount) AS total_contributed \
    FROM schedule_a_contributions \
    WHERE committee_id IN ('C00467571', 'C00618777', 'C00633792') \
        AND contributor_name LIKE '%,%' \
    GROUP BY INITCAP(SPLIT_PART(contributor_name, ',', 1)), contributor_name, contr
    HAVING SUM(contribution_receipt_amount) > 1000 \
    ORDER BY last_name, total_contributed DESC;

family_contributions_df = family_contributions
```

* postgresql://postgres:***@localhost/political_finance_data
266 rows affected.

Horizontal Bar Chart showing donations from top 10 families

```
In [253... # Group by last_name, aggregate total_contributed and collect unique states
agg = (
    family_contributions_df
    .groupby('last_name')
    .agg(
        total_contributed=('total_contributed', 'sum')
    )
    .sort_values('total_contributed', ascending=False)
    .head(10)
)

# Create Labels: "LastName (State1, State2, ...)"
labels = [f"{name}" for name, row in agg.iterrows()]

# Plot horizontal bar chart
plt.figure(figsize=(10, 6))
sns.barplot(
    x=agg['total_contributed'],
    y=labels,
    palette="viridis"
```

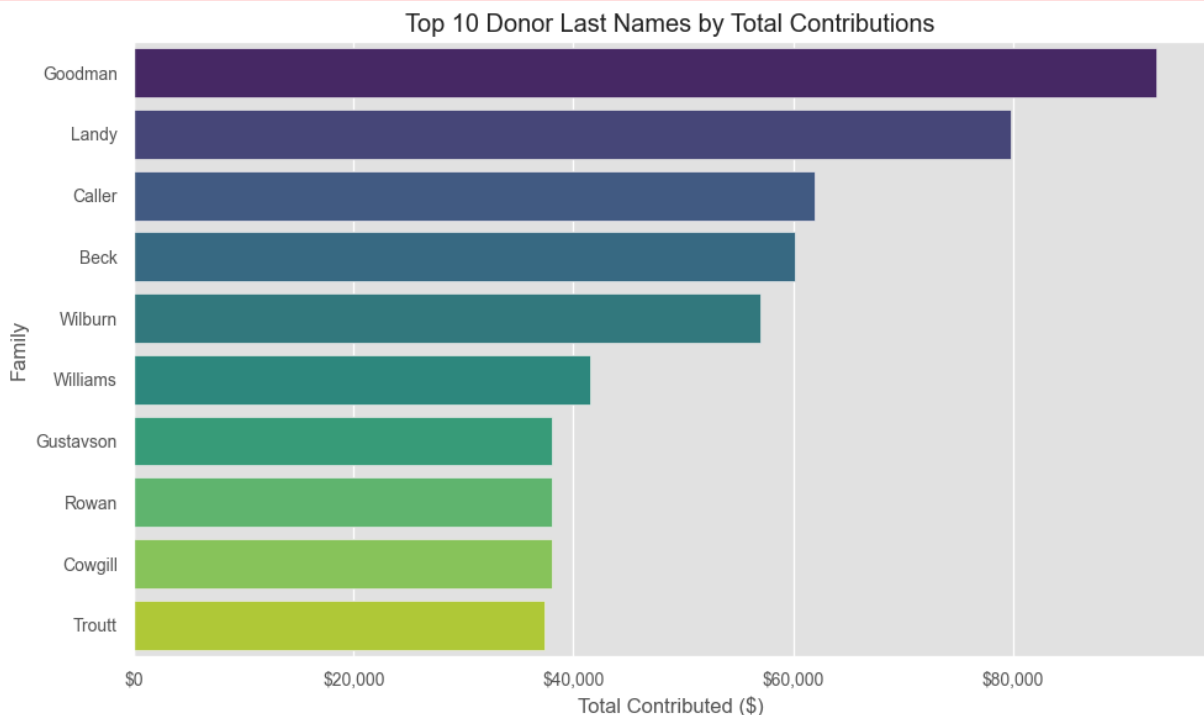
```

)
plt.xlabel("Total Contributed ($)")
plt.ylabel("Family")
plt.title("Top 10 Donor Last Names by Total Contributions")
plt.gca().xaxis.set_major_formatter(ticker.FuncFormatter(lambda x, _: f'${x:,.0f}'))
plt.tight_layout()
plt.show()

```

C:\Users\rasho\AppData\Local\Temp\ipykernel_124420\472886051.py:17: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.



A Spoke-and-Wheel Last Name Network Chart

In [231...

```

# --- STEP 2: Group donors by last name ---
clusters = defaultdict(list)
totals = defaultdict(float)

for _, row in family_contributions_df.iterrows():
    last = row['last_name']
    full = row['contributor_name']
    amt = float(row['total_contributed'])
    clusters[last].append(full)
    totals[last] += amt

# --- STEP 3: Filter to coordinated families ---
max_individual_donation = 3300 # Senate individual contribution limit
filtered_clusters = {

```

```

    k: v for k, v in clusters.items()
    if totals[k] > max_individual_donation and len(v) > 1
}

# --- STEP 4: Build the graph ---
G = nx.Graph()
family_graphs = {}
family_nodes = set()

# Sort families by total contributions
sorted_clusters = sorted(
    filtered_clusters.items(),
    key=lambda kv: totals[kv[0]],
    reverse=True
)

for family, donors in sorted_clusters:
    total = totals[family]
    family_label = f"{family} (${total/1000:.1f}K)"
    subG = nx.Graph()
    subG.add_node(family_label)
    family_nodes.add(family_label)

    for donor in donors:
        subG.add_node(donor)
        subG.add_edge(family_label, donor)

    family_graphs[family_label] = subG

# Combine all into one graph
for subG in family_graphs.values():
    G = nx.compose(G, subG)

# --- STEP 5: Layout in grid ---
pos = {}
cols = 5
padding = 3
i = 0

for family_label, subG in family_graphs.items():
    local_pos = nx.spring_layout(subG, seed=42, k=0.5) # tighter layout
    row, col = divmod(i, cols)
    x_offset = col * padding
    y_offset = -row * padding
    i += 1

    for node, (x, y) in local_pos.items():
        pos[node] = (x + x_offset, y + y_offset)

# --- STEP 6: Draw the network ---
plt.figure(figsize=(22, 18))

family_node_list = list(family_nodes)
donor_node_list = [n for n in G.nodes if n not in family_nodes]

nx.draw_networkx_nodes(G, pos, nodelist=donor_node_list, node_color='skyblue', node

```

```

nx.draw_networkx_nodes(G, pos, nodelist=family_node_list, node_color='orange', node
nx.draw_networkx_edges(G, pos, edge_color='gray')

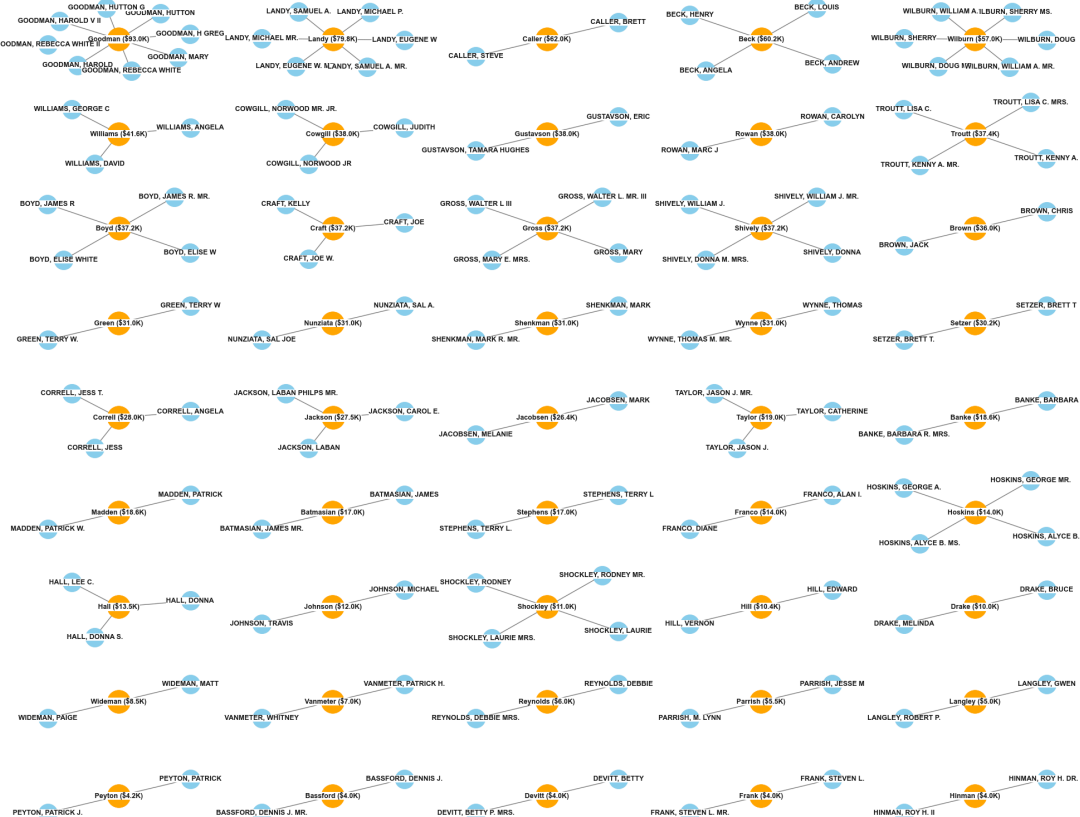
# Label formatting
labels = {}
for node in G.nodes:
    labels[node] = node

nx.draw_networkx_labels(
    G, pos, labels=labels, font_size=9,
    font_weight='bold', bbox=dict(facecolor='white', edgecolor='none', pad=0.2)
)

plt.title("Donor Network - Coordinated Family Clusters (Andy Barr)", fontsize=16)
plt.axis('off')
plt.tight_layout()
plt.show()

```

Donor Network - Coordinated Family Clusters (Andy Barr)



A Hierarchical Dendrogram to visualize relationships among donors within family groups

helps highlight suspiciously similar names (e.g., slight variations that suggest evasion of donation limits). But I don't like it...

```

In [9]: # --- Group by Last name ---
clusters = defaultdict(list)
totals = defaultdict(float)

for _, row in family_contributions.iterrows():
    last = row['last_name']
    full = row['contributor_name']
    amt = float(row['total_contributed'])
    clusters[last].append(full)
    totals[last] += amt

# --- Step 3: Top N family clusters ---
top_n = 10
top_families = sorted(totals.items(), key=lambda x: x[1], reverse=True)[:top_n]

# --- Step 4: Generate dendrograms ---
def normalized_lev_dist(a, b):
    return levenshtein_distance(a, b) / max(len(a), len(b))

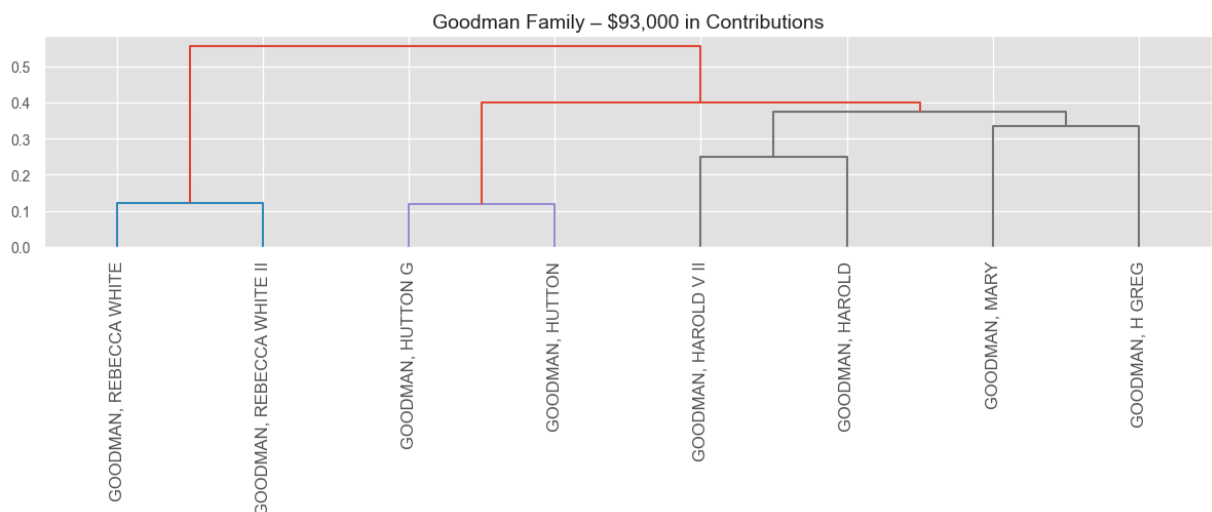
for family, total_amt in top_families:
    donors = list(set(clusters[family]))
    if len(donors) < 2:
        continue

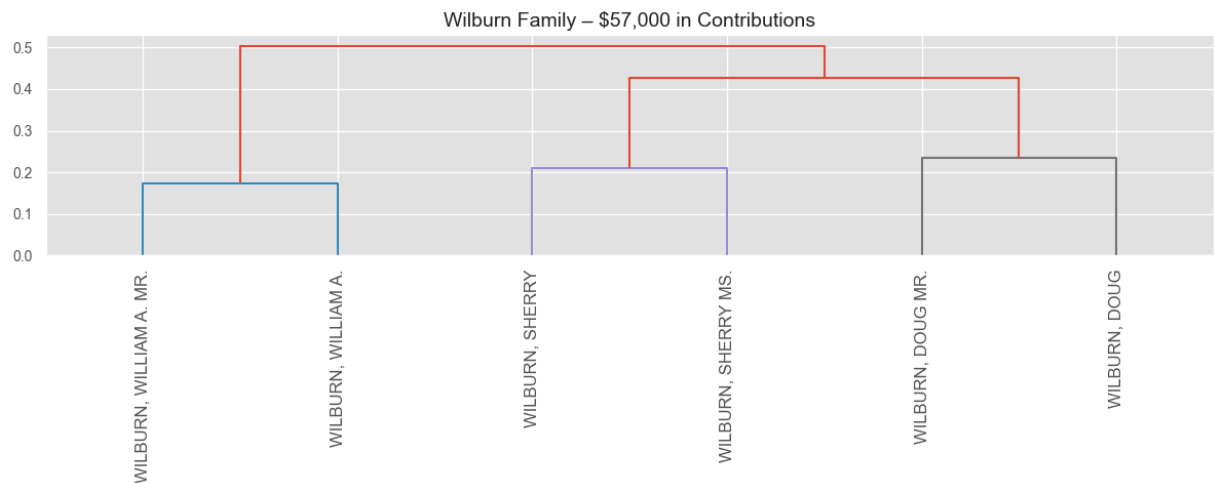
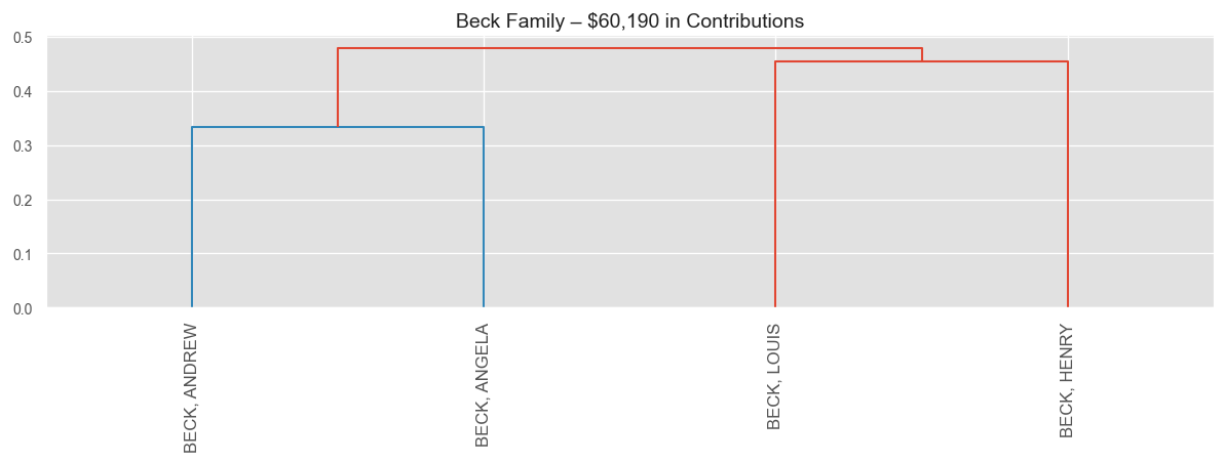
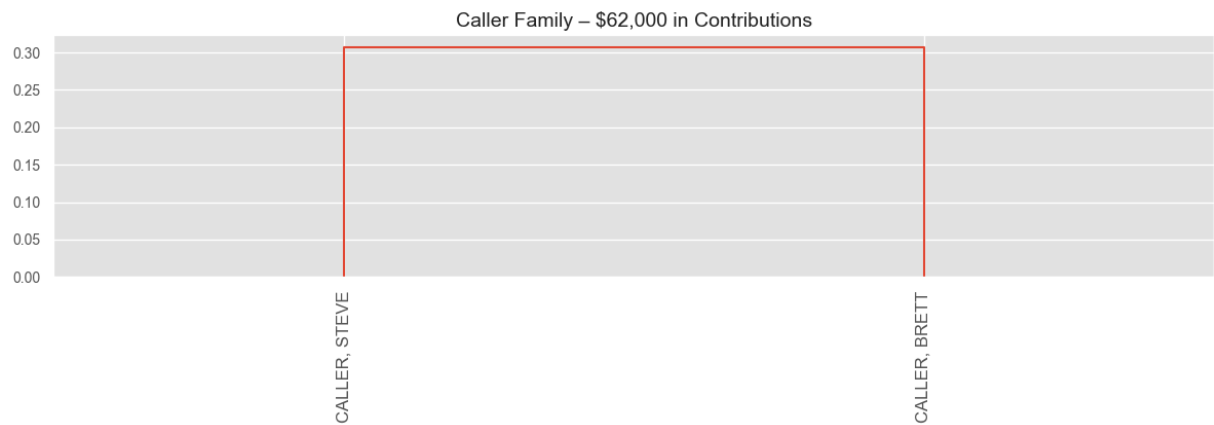
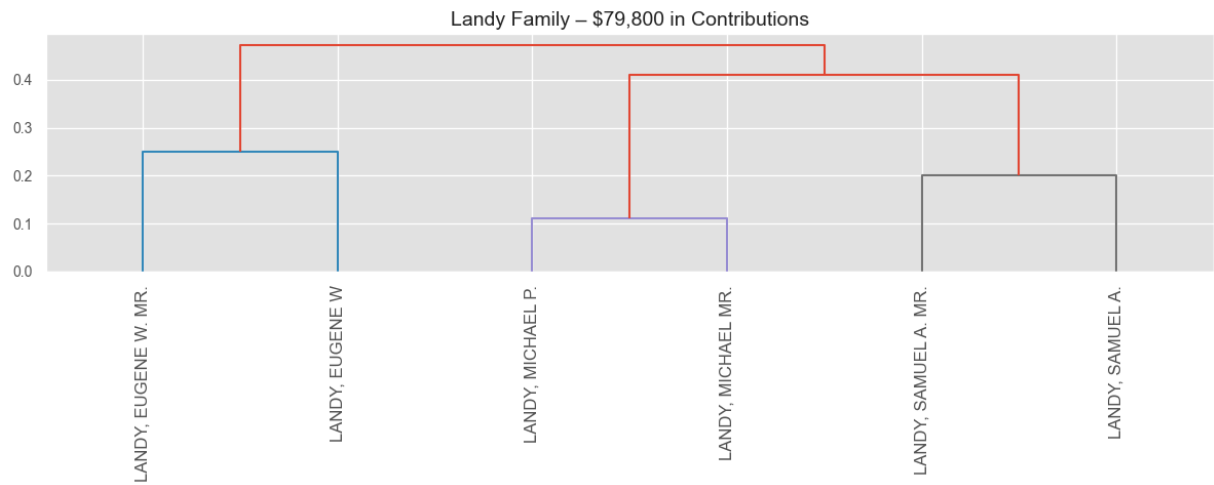
    distances = np.zeros((len(donors), len(donors)))
    for i in range(len(donors)):
        for j in range(i + 1, len(donors)):
            d = normalized_lev_dist(donors[i], donors[j])
            distances[i, j] = d
            distances[j, i] = d

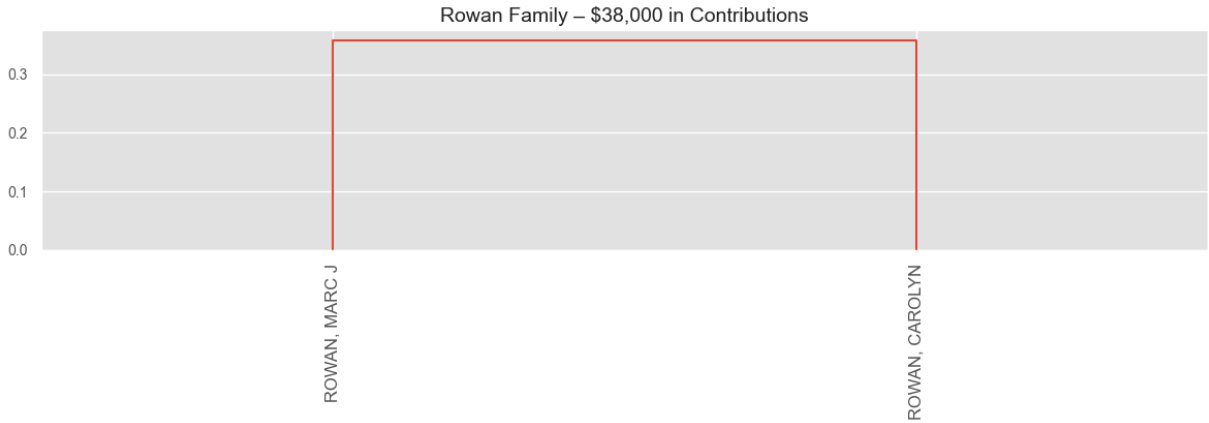
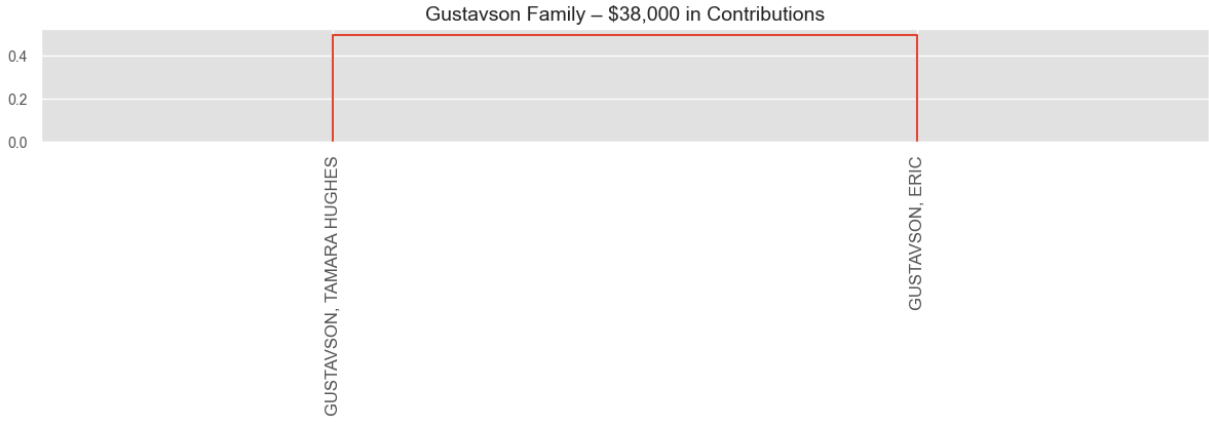
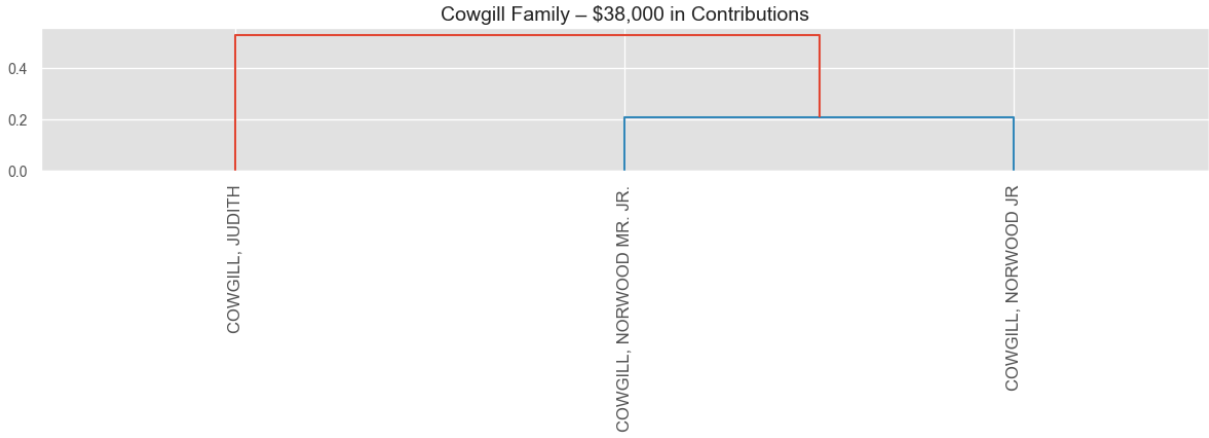
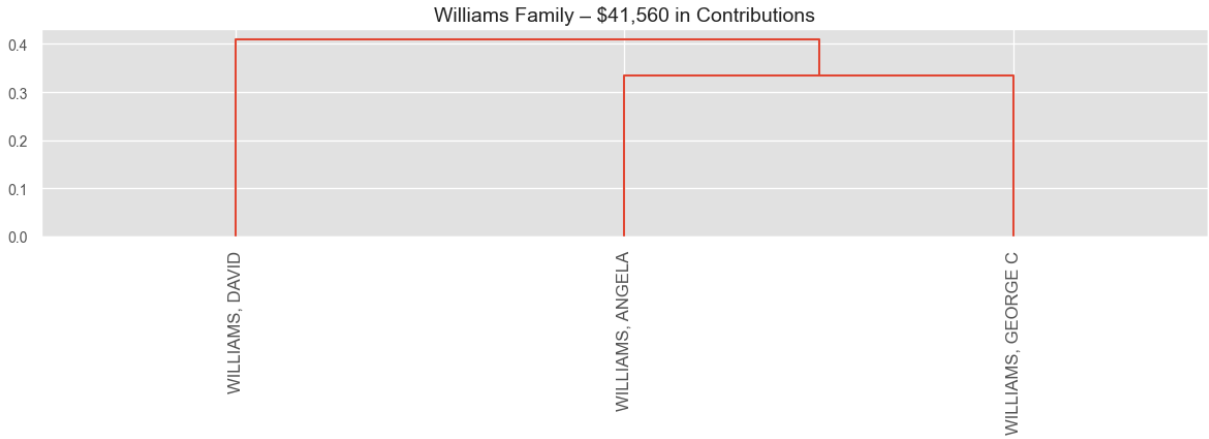
    condensed = squareform(distances)
    Z = linkage(condensed, method='average')

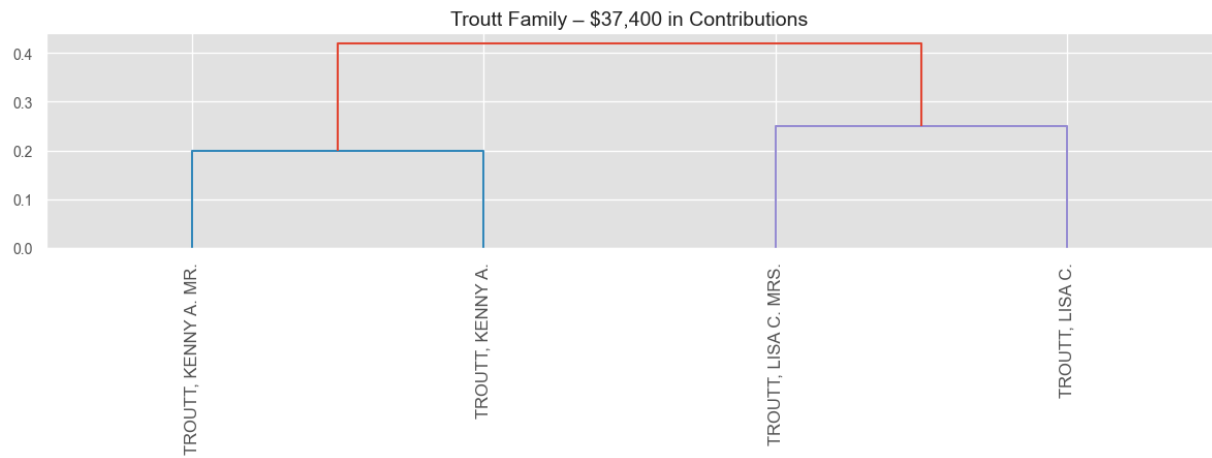
    plt.figure(figsize=(12, 4 + len(donors)*0.15))
    dendrogram(Z, labels=donors, leaf_rotation=90)
    plt.title(f"{family} Family - ${total_amt:,.0f} in Contributions", fontsize=14)
    plt.tight_layout()
    plt.show()

```









ASCII style chart

Much cleaner and more compact way of showing family groups

In [232...

```
from collections import defaultdict

# --- Print each family cluster as a text tree ---
for family, donors in sorted(filtered_clusters.items(), key=lambda x: -totals[x[0]])
    total_amt = totals[family]
    print(f"{family} (${total_amt/1000:.1f}K)")

    for i, donor in enumerate(donors):
        connector = "└─" if i == len(donors) - 1 else "├─"
        print(f"{connector} {donor}")

    print() # empty line between families
```

Goodman (\$93.0K)

- | GOODMAN, H GREG
- | GOODMAN, HAROLD
- | GOODMAN, HAROLD V II
- | GOODMAN, HUTTON
- | GOODMAN, HUTTON G
- | GOODMAN, MARY
- | GOODMAN, REBECCA WHITE
- | GOODMAN, REBECCA WHITE II

Landy (\$79.8K)

- | LANDY, EUGENE W
- | LANDY, EUGENE W. MR.
- | LANDY, MICHAEL MR.
- | LANDY, MICHAEL P.
- | LANDY, SAMUEL A.
- | LANDY, SAMUEL A. MR.

Caller (\$62.0K)

- | CALLER, BRETT
- | CALLER, STEVE

Beck (\$60.2K)

- | BECK, ANDREW
- | BECK, ANGELA
- | BECK, HENRY
- | BECK, LOUIS

Wilburn (\$57.0K)

- | WILBURN, DOUG
- | WILBURN, DOUG MR.
- | WILBURN, SHERRY
- | WILBURN, SHERRY MS.
- | WILBURN, WILLIAM A.
- | WILBURN, WILLIAM A. MR.

Williams (\$41.6K)

- | WILLIAMS, ANGELA
- | WILLIAMS, DAVID
- | WILLIAMS, GEORGE C

Cowgill (\$38.0K)

- | COWGILL, JUDITH
- | COWGILL, NORWOOD JR
- | COWGILL, NORWOOD MR. JR.

Gustavson (\$38.0K)

- | GUSTAVSON, ERIC
- | GUSTAVSON, TAMARA HUGHES

Rowan (\$38.0K)

- | ROWAN, CAROLYN
- | ROWAN, MARC J

Troutt (\$37.4K)

- | TROUTT, KENNY A.

- |— TROUTT, KENNY A. MR.
- |— TROUTT, LISA C.
- |— TROUTT, LISA C. MRS.

Boyd (\$37.2K)

- |— BOYD, ELISE W
- |— BOYD, ELISE WHITE
- |— BOYD, JAMES R
- |— BOYD, JAMES R. MR.

Craft (\$37.2K)

- |— CRAFT, JOE
- |— CRAFT, JOE W.
- |— CRAFT, KELLY

Gross (\$37.2K)

- |— GROSS, MARY
- |— GROSS, MARY E. MRS.
- |— GROSS, WALTER L III
- |— GROSS, WALTER L. MR. III

Shively (\$37.2K)

- |— SHIVELY, DONNA
- |— SHIVELY, DONNA M. MRS.
- |— SHIVELY, WILLIAM J.
- |— SHIVELY, WILLIAM J. MR.

Brown (\$36.0K)

- |— BROWN, CHRIS
- |— BROWN, JACK

Green (\$31.0K)

- |— GREEN, TERRY W
- |— GREEN, TERRY W.

Nunziata (\$31.0K)

- |— NUNZIATA, SAL A.
- |— NUNZIATA, SAL JOE

Shenkman (\$31.0K)

- |— SHENKMAN, MARK
- |— SHENKMAN, MARK R. MR.

Wynne (\$31.0K)

- |— WYNNE, THOMAS
- |— WYNNE, THOMAS M. MR.

Setzer (\$30.2K)

- |— SETZER, BRETT T
- |— SETZER, BRETT T.

Correll (\$28.0K)

- |— CORRELL, ANGELA
- |— CORRELL, JESS
- |— CORRELL, JESS T.

Jackson (\$27.5K)

- |— JACKSON, CAROL E.
- |— JACKSON, LABAN
- |— JACKSON, LABAN PHILPS MR.

Jacobsen (\$26.4K)

- |— JACOBSEN, MARK
- |— JACOBSEN, MELANIE

Taylor (\$19.0K)

- |— TAYLOR, CATHERINE
- |— TAYLOR, JASON J.
- |— TAYLOR, JASON J. MR.

Banke (\$18.6K)

- |— BANKE, BARBARA
- |— BANKE, BARBARA R. MRS.

Madden (\$18.6K)

- |— MADDEN, PATRICK
- |— MADDEN, PATRICK W.

Batmasian (\$17.0K)

- |— BATMASIAN, JAMES
- |— BATMASIAN, JAMES MR.

Stephens (\$17.0K)

- |— STEPHENS, TERRY L
- |— STEPHENS, TERRY L.

Franco (\$14.0K)

- |— FRANCO, ALAN I.
- |— FRANCO, DIANE

Hoskins (\$14.0K)

- |— HOSKINS, ALYCE B.
- |— HOSKINS, ALYCE B. MS.
- |— HOSKINS, GEORGE A.
- |— HOSKINS, GEORGE MR.

Hall (\$13.5K)

- |— HALL, DONNA
- |— HALL, DONNA S.
- |— HALL, LEE C.

Johnson (\$12.0K)

- |— JOHNSON, MICHAEL
- |— JOHNSON, TRAVIS

Shockley (\$11.0K)

- |— SHOCKLEY, LAURIE
- |— SHOCKLEY, LAURIE MRS.
- |— SHOCKLEY, RODNEY
- |— SHOCKLEY, RODNEY MR.

Hill (\$10.4K)

- └ HILL, EDWARD
- └ HILL, VERNON

Drake (\$10.0K)

- └ DRAKE, BRUCE
- └ DRAKE, MELINDA

Wideman (\$8.5K)

- └ WIDEMAN, MATT
- └ WIDEMAN, PAIGE

Vanmeter (\$7.0K)

- └ VANMETER, PATRICK H.
- └ VANMETER, WHITNEY

Reynolds (\$6.0K)

- └ REYNOLDS, DEBBIE
- └ REYNOLDS, DEBBIE MRS.

Parrish (\$5.5K)

- └ PARRISH, JESSE M
- └ PARRISH, M. LYNN

Langley (\$5.0K)

- └ LANGLEY, GWEN
- └ LANGLEY, ROBERT P.

Peyton (\$4.2K)

- └ PEYTON, PATRICK
- └ PEYTON, PATRICK J.

Bassford (\$4.0K)

- └ BASSFORD, DENNIS J.
- └ BASSFORD, DENNIS J. MR.

Devitt (\$4.0K)

- └ DEVITT, BETTY
- └ DEVITT, BETTY P. MRS.

Frank (\$4.0K)

- └ FRANK, STEVEN L.
- └ FRANK, STEVEN L. MR.

Hinman (\$4.0K)

- └ HINMAN, ROY H. DR. II
- └ HINMAN, ROY H. II



Employer-Based Network

clustering donors based on their reported employer names often reveals potential bundling or coordinated fundraising through businesses, law firms, or other organizations. But it doesn't look right, the way a few yellow balls are connected in pairs. why?

```
In [29]: # -- Query contributions by employer
employer_contributions = %sql \
SELECT \
    TRIM(UPPER(contributor_occupation)) AS occupation, \
    TRIM(UPPER(contributor_employer)) AS employer, \
    contributor_name, \
    SUM(contribution_receipt_amount) AS total_contributed \
FROM schedule_a_contributions \
WHERE committee_id IN ('C00467571', 'C00618777', 'C00633792') \
    AND contributor_name LIKE '%,%' \
    AND contributor_employer IS NOT NULL \
    AND contributor_employer != '' \
GROUP BY contributor_employer, contributor_occupation, contributor_name \
HAVING SUM(contribution_receipt_amount) > 100 \
ORDER BY contributor_employer;
```

* postgresql://postgres:***@localhost/political_finance_data
434 rows affected.

```
In [34]: # Convert SQL result to DataFrame
df = employer_contributions

excluded_employers = {"self-employed", "self employed", "none", "n/a", "self", "not"}
filtered_df = df[
    ~df["employer"].str.lower().isin(excluded_employers)
]

# Build clusters by employer
clusters = defaultdict(list)
totals = defaultdict(float)

for _, row in filtered_df.iterrows():
    employer = row['employer']
    name = row['contributor_name']
    amount = float(row['total_contributed'])
    clusters[employer].append(name)
    totals[employer] += amount

# Filter clusters with more than 1 donor and high total contribution
filtered = {
    emp: donors for emp, donors in clusters.items()
    if len(donors) > 1 and totals[emp] > 5000
}

# --- Build graph ---
G = nx.Graph()
for employer, donors in filtered.items():
    label = f"{employer} (${totals[employer]/1000:.1f}K)"
    G.add_node(label, group=employer)

    for donor in donors:
        G.add_edge(label, donor)

# --- Layout and plot ---
plt.figure(figsize=(20, 14))
pos = nx.spring_layout(G, seed=42, k=2 / math.sqrt(len(G.nodes)))
```

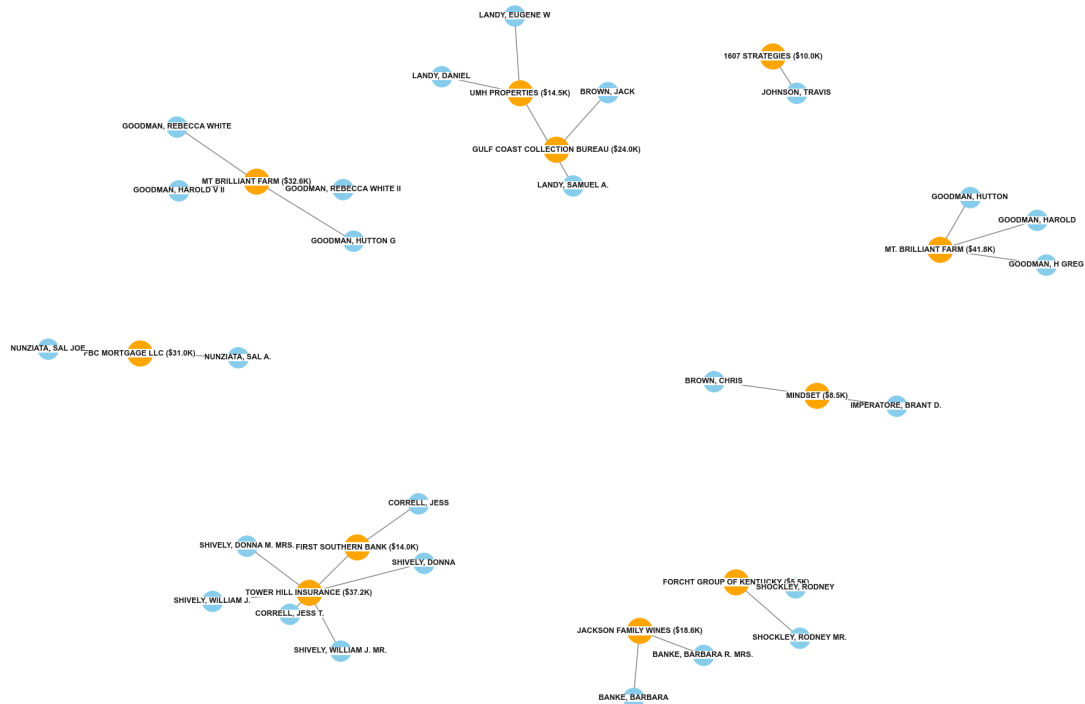


```
# Split family vs donor nodes
hub_nodes = [n for n in G.nodes if '($' in n]
donor_nodes = [n for n in G.nodes if n not in hub_nodes]
nx.draw_networkx_nodes(G, pos, nodelist=hub_nodes, node_color='orange', node_size=9)
nx.draw_networkx_nodes(G, pos, nodelist=donor_nodes, node_color='skyblue', node_size=9)
nx.draw_networkx_edges(G, pos, edge_color='gray')

nx.draw_networkx_labels(
    G, pos, font_size=9, font_weight='bold',
    bbox=dict(facecolor='white', edgecolor='none', pad=0.2)
)

plt.title("Donor Network - Employer-Based Clusters (Andy Barr Committees)", font_size=12)
plt.axis('off')
plt.tight_layout()
plt.show()
```

Donor Network – Employer-Based Clusters (Andy Barr Committees)



🧱 LLC or Entity Bundling

🏛️ Contribution Rules for LLCs and Entities

- **LLCs Treated as Partnerships:** If an LLC is classified as a partnership for tax purposes, it may contribute to federal candidates. These contributions count against both the partnership's limit and the individual partners' limits, based on their share of ownership.

At the time of the contribution, the LLC must inform the recipient committee of its tax status and how the contribution should be attributed .

- **LLCs Treated as Corporations:** If an LLC is taxed as a corporation, it is prohibited from making contributions to federal candidates .
- **Corporations and Labor Organizations:** These entities are generally prohibited from making contributions to federal candidates .

Contribution Limits

For entities permitted to contribute (e.g., partnerships), the contribution limits are as follows:

- **Per Election Limit:** \$3,300 per election to a federal candidate.

This means an entity can contribute up to 3,300 *for the primary election and another 3,300* for the general election, totaling \$6,600 if both elections are contested.

Compliance Considerations

- **Attribution:** When an LLC contributes, it must specify how the contribution is attributed among its members.
- **Disclosure:** The recipient committee must disclose the contribution and its attribution in their FEC filings.
- **Prohibited Contributions:** Contributions from entities not permitted to donate (e.g., corporations) must be refused or refunded.

Understanding these rules is crucial to ensure compliance with federal campaign finance laws.

In [127...

```
%%sql
-- Step 1: Create a temporary table with contributions from entity donors
DROP TABLE IF EXISTS entity_donors;
CREATE TEMP TABLE entity_donors AS
SELECT
    contributor_name,
    contributor_state,
    contribution_receipt_amount
FROM
    schedule_a_contributions
WHERE
    committee_id IN ('C00467571', 'C00618777', 'C00633792')
-- Filter for contributor names containing common entity identifiers
AND contributor_name ILIKE ANY (ARRAY[
    '% LLC%', '% INC%', '% CORP%', '% CO %', '%COMPANY%',
    '% LTD%', '% LP%', '% LLP%', '% PC%', '%PLC%', '%PLLC%',
```

```

        '%TRUST%', '%ASSOCIATES%', '%GROUP%', '%FUND%',
        '%PARTNERS%', '%ENTERPRISES%', '%HOLDINGS%'
    ]);

-- Step 2: Aggregate contributions by entity donor
DROP TABLE IF EXISTS entity_clusters;
CREATE TEMP TABLE entity_clusters AS
SELECT
    contributor_name,
    contributor_state,
    COUNT(*) AS num_contributions,
    SUM(contribution_receipt_amount) AS total_contributed,
    SUM(contribution_receipt_amount) / COUNT(*) AS avg_contribution_per_donor
FROM
    entity_donors
GROUP BY
    contributor_name,
    contributor_state;

```

```

* postgresql://postgres:***@localhost/political_finance_data
Done.
128 rows affected.
Done.
93 rows affected.

```

Out[127... —

```

In [128... # Step 3: Select clusters where average contribution per donor exceeds $3,300
entity_contributions = %sql SELECT \
    contributor_name, \
    contributor_state, \
    num_contributions, \
    total_contributed, \
    avg_contribution_per_donor \
FROM \
    entity_clusters \
WHERE \
    avg_contribution_per_donor > 3300 \
ORDER BY \
    total_contributed DESC;

```

```

* postgresql://postgres:***@localhost/political_finance_data
52 rows affected.

```

Entity Contributions and the states they originate from

In [129... entity_contributions

Out[129...

	contributor_name	contributor_state	num_contributions	total_contributed	avg_cont
0	FORTUNE TRADE CENTER, LLC	KY	1	24000.00	24000.00
1	CALLER PROPERTIES, LLC	KY	1	24000.00	24000.00
2	POLO 1, LLC	KY	1	11600.00	11600.00
3	UBS AMERICAS INC. POLITICAL ACTION COMMITTEE (...)	CT	2	10000.00	5000.00
4	ALLY FINANCIAL INC. ADVOCACY PAC	DC	2	10000.00	5000.00
5	NOMURA HOLDING AMERICA INC. PAC	DC	2	10000.00	5000.00
6	MANAGED FUNDS ASSOCIATION PAC	DC	2	10000.00	5000.00
7	BARCLAYS GROUP US INC. PAC	DC	2	10000.00	5000.00
8	MARATHON PETROLEUM CORPORATION EMPLOYEES PAC	OH	2	10000.00	5000.00
9	THE GOLDMAN SACHS GROUP, INC. POLITICAL ACTION...	DC	2	10000.00	5000.00
10	INVESTMENT COMPANY INSTITUTE PAC	DC	2	10000.00	5000.00
11	KPMG PARTNERS/PRINCIPALS AND EMPLOYEES PAC	DC	2	10000.00	5000.00
12	INDEPENDENT INSURANCE AGENTS & BROKERS OF AMER...	DC	2	10000.00	5000.00
13	TRUIST FINANCIAL CORPORATION FEDERAL PAC (FORM...	VA	2	10000.00	5000.00
14	ELEVANCE HEALTH, INC. PAC (ELEVANCE HEALTH PAC)	DC	2	10000.00	5000.00
15	SAZERAC COMPANY, INC. PAC (SAZERACPAC)	DC	3	10000.00	3333.33

	contributor_name	contributor_state	num_contributions	total_contributed	avg_cont
16	BEAM SUNTORY INC PAC	DC	2	10000.00	5000
17	FEDERAL BIPARTISAN VOLUNTARY PUBLIC AFFAIRS CO...	DC	2	10000.00	5000
18	TOYOTA MOTOR NORTH AMERICA, INC POLITICAL ACTI...	DC	2	8500.00	4250
19	POLO 1 LLC	KY	2	7000.00	3500
20	CALLER PROPERTIES LLC	KY	2	7000.00	3500
21	FIRST HORIZON NATIONAL CORPORATION FEDERAL PAC	TN	2	7000.00	3500
22	FORTUNE TRADE CENTER LLC	KY	2	7000.00	3500
23	LIBERTY MUTUAL INSURANCE COMPANY PAC	MA	1	5000.00	5000
24	ZURICH HOLDING COMPANY OF AMERICA COMMITTEE FO...	DC	1	5000.00	5000
25	PRINCIPAL LIFE INSURANCE COMPANY PAC	IA	1	5000.00	5000
26	ROCK HOLDINGS INC. PAC	MI	1	5000.00	5000
27	GENERAL ELECTRIC COMPANY POLITICAL ACTION COMM...	DC	1	5000.00	5000
28	ALTRIA GROUP, INC. POLITICAL ACTION COMMITTEE ...	DC	1	5000.00	5000
29	FEDEX CORPORATION PAC	TN	1	5000.00	5000
30	METLIFE INC. EMPLOYEES' POLITICAL PARTICIPATIO...	NY	1	5000.00	5000

	contributor_name	contributor_state	num_contributions	total_contributed	avg_cont
31	THE BANK OF NEW YORK MELLON CORPORATION PAC	PA	1	5000.00	5000
32	BLACKROCK CAPITAL MANAGEMENT INC. PAC	NY	1	5000.00	5000
33	VALERO ENERGY CORPORATION PAC	TX	1	5000.00	5000
34	REGIONS FINANCIAL CORPORATION PAC	DC	1	5000.00	5000
35	AMERICA'S CREDIT UNIONS PAC OF CREDIT UNION NA...	DC	1	5000.00	5000
36	CME GROUP INC. PAC	IL	1	5000.00	5000
37	CAPITAL ONE FINANCIAL CORP. ASSOC. POLITICAL FUND	VA	1	5000.00	5000
38	THE CAPITAL GROUP COMPANIES INC. PAC	CA	1	5000.00	5000
39	BROWN-FORMAN CORPORATION NON- PARTISAN COMMITTE...	KY	1	5000.00	5000
40	UNITEDHEALTH GROUP INCORPORATED PAC (UNITED FO...	DC	1	5000.00	5000
41	BMO FINANCIAL CORP GOOD GOVERNANCE FUND: FEDER...	IL	1	5000.00	5000
42	HSBC TECHNOLOGY AND SERVICES USA INC. FEDERAL,...	DC	1	5000.00	5000
43	SOUTHERN NUCLEAR OPERATING COMPANY, INC. PAC	AL	1	5000.00	5000
44	LPL FINANCIAL LLC POLITICAL ACTION COMMITTEE (...)	CA	1	5000.00	5000
45	KOCH INDUSTRIES, INC. POLITICAL ACTION COMMITT...	KS	1	5000.00	5000

	contributor_name	contributor_state	num_contributions	total_contributed	avg_cont
46	WELLS FARGO AND COMPANY EMPLOYEE PAC	MN	1	5000.00	5000
47	CALPORTLAND COMPANY PAC (CPCC- PAC)	CA	1	5000.00	5000
48	ALLSTATE INSURANCE COMPANY PAC	IL	1	5000.00	5000
49	BANK OF AMERICA CORPORATION FEDERAL PAC	DC	1	5000.00	5000
50	NATIONAL SHOOTING SPORTS FOUNDATION, INC. PAC ...	DC	1	5000.00	5000
51	FMR LLC PAC - FEDERAL (FIDELITY PAC)	MA	1	4000.00	4000

Shared Address Clustering

Benefits of full address clustering:

- Detects household-level coordination: If multiple contributions come from the same address but different names, it may indicate a family or coordinated group bundling donations.
- Captures entity or shell activity: Some donation schemes involve creating multiple entities registered to the same address (e.g., law firms, LLCs, or PO boxes).
- More precise than ZIP code: ZIP codes can span entire neighborhoods or cities; full addresses can narrow down coordination to a building or apartment.

Use cases:

- Donor networks: Identify clusters of donations from a single building (e.g., luxury condos often used for donor aggregation).
- Straw donations: Look for unusually high aggregate donations from one household.
- PAC and shell LLCs: Spot contributions from multiple orgs with the same registered agent address.

In [278...

```
%%sql
DROP TABLE IF EXISTS full_address_clusters;
CREATE TABLE full_address_clusters AS
SELECT
    TRIM(UPPER(CONCAT_WS(' ',
        contributor_street_1,
        contributor_street_2,
        contributor_city,
        contributor_state,
        LEFT(contributor_zip, 5)
    ))) AS full_address,
    COUNT(*) AS donation_count,
    SUM(contribution_receipt_amount) AS total_donated,
    ARRAY_AGG(DISTINCT contributor_name) AS unique_donors
FROM schedule_a_contributions
WHERE contributor_zip IS NOT NULL
    AND contributor_street_1 IS NOT NULL
    AND committee_id IN ('C00467571', 'C00618777', 'C00633792')
GROUP BY full_address
HAVING COUNT(*) > 5
ORDER BY total_donated DESC;
```

* postgresql://postgres:***@localhost/political_finance_data
 Done.
 66 rows affected.

Out[278... —

Explanation:

- clustered_contributions: This is your table containing contribution data with a cluster_id assigned to each contributor.
- contributor_id: A unique identifier for each individual contributor. Ensure this field accurately represents individual donors.
- contribution_amount: The amount contributed by each individual.
- cluster_id: An identifier representing a group of contributors clustered based on criteria like last name or ZIP code.

Steps:

- Aggregate Data: The cluster_contributions CTE (Common Table Expression) calculates:
 - num_individuals: Number of unique contributors in each cluster.
 - total_contributions: Total contributions received from each cluster.
 - avg_contribution_per_individual: Average contribution per individual in the cluster.
- Filter Clusters: The main query selects clusters where the average contribution per individual exceeds the legal limit of \$3,500 per election.

3. Order Results: The results are ordered in descending order of average contribution per individual to highlight the most significant cases.

Note: Ensure that the contributor_id field accurately represents individual donors to avoid skewed averages due to duplicate or misidentified contributors.

```
In [93]: df_address_clusters =%sql SELECT * \
FROM full_address_clusters \
WHERE (total_donated / array_length(unique_donors, 1)) > 3300 \
ORDER BY total_donated DESC;
df_address_clusters
```

```
* postgresql://postgres:***@localhost/political_finance_data
17 rows affected.
```

Out[93]:

	full_address	donation_count	total_donated	unique_donors
0	PO BOX 9891 ARLINGTON VA 22219	1179	491150.50	[WINRED]
1	840 E HIGH ST LEXINGTON KY 40502	14	124000.00	[CALLER PROPERTIES LLC, CALLER PROPERTIES, LLC...
2	3314 HUFFMAN MILL PIKE LEXINGTON KY 40511	15	93000.00	[GOODMAN, H GREG, GOODMAN, HAROLD, GOODMAN, HA...
3	4701 GRANDVIEW CT MCFARLAND WI 53558	6	61000.00	[BROWN, JACK, HAAG, TIM]
4	2401 SIR BARTON WAY LEXINGTON KY 40509	7	54386.61	[FORCHT BANK]
5	3750 PARIS PIKE LEXINGTON KY 40511	8	38793.38	[BECK, ANDREW, BECK, ANGELA]
6	884 IRON WORKS PIKE LEXINGTON KY 40511	6	38000.00	[GUSTAVSON, ERIC, GUSTAVSON, TAMARA HUGHES]
7	1999 RICHMOND RD STE 200 LEXINGTON KY 40502	6	38000.00	[COWGILL, JUDITH, COWGILL, NORWOOD JR, COWGILL...
8	3430 CHARLES ELGIN RD EUBANK KY 42567	6	38000.00	[WILBURN, SHERRY, WILBURN, SHERRY MS., WILBURN...
9	PO BOX 12128 LEXINGTON KY 40580	6	37200.00	[MADDEN, PATRICK, MADDEN, PATRICK W., POLO 1 L...
10	559 S FORBES RD LEXINGTON KY 40504	6	37200.00	[GROSS, MARY, GROSS, MARY E. MRS., GROSS, WALT...
11	1717 S BOULDER AVE STE 400 TULSA OK 74119	6	37200.00	[CRAFT, JOE, CRAFT, JOE W., CRAFT, KELLY]
12	251 H ST NW WASHINGTON DC 20001	10	27000.00	[AMERICAN ISRAEL PUBLIC AFFAIRS COMMITTEE PAC]
13	1400 DIXIANA DOMINO RD LEXINGTON KY 40511	6	14000.00	[SHIVELY, DONNA, SHIVELY, WILLIAM J.]
14	1545 N OCEAN WAY PALM BEACH FL 33480	14	8559.50	[WILLIAMS, ANGELA]
15	386 W TROXELL RD OAK HARBOR WA 98277	15	4508.70	[PARKS, JENNIE]
16	262 E MAIN ST MOORESTOWN NJ 08057	69	3550.00	[HILL, VERNON]

Explanation:

- `total_contributed`: Represents the sum of contributions within each cluster.
- `individual_count`: Denotes the number of unique individuals in each cluster.
- The `WHERE` clause filters clusters where the average contribution per individual exceeds \$3,300.
- `ORDER BY total_contributed DESC`: Sorts the results in descending order based on total contributions, highlighting the most significant clusters first.

Note:

- Ensure that your `full_address_clusters` temporary table is defined and populated within the same session, as temporary tables are session-scoped and will not persist across different sessions.
- If you encounter issues accessing the temporary table, consider verifying its creation and scope.



Out-of-State Influence

- Title: "Contributions to Andy's Committees from out-of-state"
- Chart Type: Horizontal bar chart or map visualization

In [210...

```
state_contributions = %sql SELECT \
    sa.contributor_state, \
    ROUND(SUM(sa.contribution_receipt_amount) / 1000, 1) as total_k\
FROM schedule_a_contributions sa \
JOIN committees c ON sa.committee_id = c.committee_id \
WHERE sa.committee_id in ('C00467571', 'C00618777', 'C00633792') \
GROUP BY sa.contributor_state \
ORDER BY SUM(sa.contribution_receipt_amount) DESC;
```

```
* postgresql://postgres:***@localhost/political_finance_data
42 rows affected.
```

In [211...

```
state_contributions
```

Out[211...

	contributor_state	total_k
0	KY	996.0
1	GA	589.3
2	VA	568.1
3	DC	468.7
4	FL	298.7
5	NY	127.8
6	IL	114.7
7	TX	112.0
8	CT	92.9
9	NJ	79.2
10	CA	77.0
11	WI	64.8
12	OK	38.3
13	SC	32.6
14	OH	27.1
15	PA	25.0
16	TN	22.4
17	MD	21.3
18	MI	17.8
19	NC	15.2
20	LA	14.5
21	WA	13.3
22	PR	13.0
23	MA	11.5
24	MO	8.9
25	IA	7.7
26	WV	7.0
27	MN	6.7
28	WY	6.6
29	NE	6.0

	contributor_state	total_k
30	KS	6.0
31	AL	5.4
32	AZ	5.4
33	ID	4.0
34	UT	3.5
35	IN	2.5
36	NV	1.8
37	SD	0.9
38	CO	0.4
39	ME	0.2
40	NH	0.2
41	MS	0.1

In [226...

```
import plotly.express as px

# Rename columns for clarity
state_df = state_contributions.rename(columns={'contributor_state': 'state', 'total': 'total_k'})
state_df['total_k'] = pd.to_numeric(state_df['total_k'], errors='coerce')
print(state_df['total_k'].describe()) # Should now show mean, std, min, max, etc.

fig = px.choropleth(
    state_df,
    locations='state',
    locationmode="USA-states",
    color='total_k',
    color_continuous_scale="Cividis", # or "Hot", "YlOrRd"
    range_color=(0, state_df['total_k'].max()),
    scope="usa",
    labels={'total_k': 'Total Contributed ($1000s)'},
    title="Contributions to Andy Barr's Committees by State<br>(Color = More Money)"
)
fig.update_layout(coloraxis_colorbar=dict(title="Total Contributed ($1000s)"))
fig.show()
```

```
count    42.000000
mean     93.202381
std      202.404969
min        0.100000
25%        5.550000
50%       13.900000
75%       73.950000
max      996.000000
Name: total_k, dtype: float64
```

Georgia based committees

First, let's look at where committees are based, that have been getting contributions

In [166...

```
%%sql
SELECT c.name, c.state, count(*) as num_contributions, sum(contribution_receipt_amo
FROM schedule_a_contributions sa
JOIN committees c ON sa.committee_id = c.committee_id
WHERE sa.committee_id in ('C00467571', 'C00618777', 'C00633792')
group by c.name, c.state
HAVING sum(contribution_receipt_amount) > 9900
AND count(*) > 2
ORDER BY sum(contribution_receipt_amount) DESC;
```

* postgresql://postgres:***@localhost/political_finance_data
2 rows affected.

Out[166...

	name	state	num_contributions	total_contributed
0	ANDY BARR FOR SENATE, INC.	KY	2874	2577038.49
1	FRIENDS OF ANDY BARR COMMITTEE	GA	192	1337457.79

Not sure what the point of this query was...

In [160...

```
%%sql
SELECT contributor_name, contributor_state, contributor_employer, contributor_occup
FROM schedule_a_contributions
WHERE committee_id IN ('C00618777', 'C00633792')
and contributor_name NOT ILIKE '%WinRed%'
--and contributor_street_1 is null
LIMIT 40;
```

* postgresql://postgres:***@localhost/political_finance_data
40 rows affected.

Out[160...

	contributor_name	contributor_state	contributor_employer	contributor_occupation	con
0	PEYTON, PATRICK	FL	NONE	RETIRED	
1	MERCEIN, TOMMY	CT	HOMIUM	OWNER	
2	MCLAUGHLIN, THOMAS MCLAUGHLIN	FL	KELLER WILLIAMS REALTY LAKELAND	OPERATING PRINCIPAL	
3	WORKMAN, RICHARD E	IL	NEW HARBOR CAPITAL	CEO	
4	HALL, DONNA	KY	None	None	
5	GOODMAN, REBECCA WHITE II	KY	MT BRILLIANT FARM	OWNER	
6	GOODMAN, MARY	KY	None	None	
7	GOODMAN, HUTTON	KY	MT. BRILLIANT FARM	RACING MANAGER	
8	GOODMAN, HAROLD	KY	MT. BRILLIANT FARM	OWNER	
9	GOODMAN, H GREG	KY	MT. BRILLIANT FARM	OWNER	
10	CALLER, BRETT	KY	CALLER PROPERTIES, LLC	OWNER	
11	CALLER, STEVE	KY	FORTUNE TRADE CENTER, LLC	OWNER	
12	FORTUNE TRADE CENTER, LLC	KY	None	None	
13	CORRELL, JESS	KY	FIRST SOUTHERN BANK	BANKER	
14	CORRELL, ANGELA	KY	SELF-EMPLOYED	WRITER	
15	CALLER, STEVE	KY	FORTUNE TRADE CENTER, LLC	OWNER	
16	CALLER, BRETT	KY	CALLER PROPERTIES, LLC	OWNER	
17	CALLER PROPERTIES, LLC	KY	None	None	
18	KOVACS, JAMES	FL	SELF EMPLOYED	INVESTOR	
19	DAVIDSON, CHARLES	FL	WEXFORD CAPITAL	CHAIRMAN	
20	GROSS, WALTER L. MR. III	KY	None	None	

	contributor_name	contributor_state	contributor_employer	contributor_occupation	con
21	TROUTT, LISA C. MRS.	TX	NONE	HOMEMAKER	
22	TROUTT, KENNY A. MR.	TX	WINSTAR FARMS	OWNER	
23	SHIVELY, WILLIAM J. MR.	KY	TOWER HILL INSURANCE	CEO/OWNER	
24	SHIVELY, WILLIAM J. MR.	KY	TOWER HILL INSURANCE	CEO/OWNER	
25	SHIVELY, DONNA M. MRS.	KY	TOWER HILL INSURANCE	EVENT PLANNER	
26	SHIVELY, DONNA M. MRS.	KY	TOWER HILL INSURANCE	EVENT PLANNER	
27	LANDY, MICHAEL MR.	FL	NONE	RETIRED	
28	LANDY, EUGENE W. MR.	NJ	None	None	
29	GROSS, MARY E. MRS.	KY	None	None	
30	BASSFORD, DENNIS J. MR.	ID	None	None	
31	HEGY, ALBERT	NY	1ST FINANCIAL BANK USA	BANKER	
32	JOHNSON, TRAVIS	VA	1607 STRATEGIES	GOVT. AFFAIRS	
33	RIEDER, RICK	NJ	BLACKROCK	CIO	
34	LEE, PENNY	DC	FINANCIAL TECHNOLOGY ASSOCIATION	CEO	
35	TAYLOR, JASON J. MR.	KY	None	None	
36	SHOCKLEY, RODNEY MR.	KY	FORCHT GROUP OF KENTUCKY	ATTORNEY	
37	SHOCKLEY, LAURIE MRS.	KY	None	None	
38	REYNOLDS, DEBBIE MRS.	KY	None	None	
39	POLO 1, LLC	KY	None	None	

4. Opaque PAC Contributions & Shell Committees

Purpose: Identify contributors that are difficult to trace, often shell PACs or 501(c)(4)s. 📊

Chart 4: Top PAC Donors with Low Transparency

- Title: "Top Opaque PACs Contributing to Andy Barr"
- Chart Type: Bar chart

In [229...

```
%%sql
SELECT contributor_name, SUM(contribution_receipt_amount) AS total_contributed
FROM schedule_a_contributions
WHERE committee_id in ('C00467571', 'C00618777', 'C00633792')
--AND entity_type = 'COM'
AND (contributor_name ILIKE '%Action%'
      OR contributor_name ILIKE '%Alliance%'
      OR contributor_name ILIKE '%America%'
      OR contributor_name ILIKE '%Policy%') -- simple heuristics
GROUP BY contributor_name
ORDER BY total_contributed DESC
LIMIT 50;
```

```
* postgresql://postgres:***@localhost/political_finance_data
44 rows affected.
```

Out[229...

	contributor_name	total_contributed
0	AMERICAN ISRAEL PUBLIC AFFAIRS COMMITTEE PAC	27000.00
1	AMERICAN PROPERTY CASUALTY INSURANCE ASSOCIATI...	10000.00
2	THE GOLDMAN SACHS GROUP, INC. POLITICAL ACTION...	10000.00
3	AMERICAN BANKERS ASSOCIATION PAC (BANKPAC)	10000.00
4	UBS AMERICAS INC. POLITICAL ACTION COMMITTEE (...)	10000.00
5	NOMURA HOLDING AMERICA INC. PAC	10000.00
6	U.S. BANCORP FEDERAL POLITICAL ACTION COMMITTEE	10000.00
7	INDEPENDENT INSURANCE AGENTS & BROKERS OF AMER...	10000.00
8	AMERICAN INSTITUTE OF CERTIFIED PUBLIC ACCOUNT...	10000.00
9	AMERICAN FINANCIAL SERVICES ASSOCIATION PAC	10000.00
10	TOYOTA MOTOR NORTH AMERICA, INC POLITICAL ACTI...	8500.00
11	TEACHERS INSURANCE ANNUITY ASSOCIATION OF AMER...	7500.00
12	AMERICAN SOCIETY OF ANESTHESIOLOGISTS POLITICA...	7500.00
13	AMERICAN COUNCIL OF LIFE INSURERS PAC	7500.00
14	ZURICH HOLDING COMPANY OF AMERICA COMMITTEE FO...	5000.00
15	ALTRIA GROUP, INC. POLITICAL ACTION COMMITTEE ...	5000.00
16	AMERICA'S CREDIT UNIONS PAC OF CREDIT UNION NA...	5000.00
17	AMERICAN ACADEMY OF DERMATOLOGY ASSOCIATION PAC	5000.00
18	AMERICAN DENTAL ASSOCIATION PAC	5000.00
19	AMERICAN INVESTMENT COUNCIL PAC - AIC PAC	5000.00
20	AMERICAN LAND TITLE ASSOCIATION TITLE INDUSTRY...	5000.00
21	BANK OF AMERICA CORPORATION FEDERAL PAC	5000.00
22	BUILDING AMERICA'S REPUBLICAN REPRESENTATION PAC	5000.00
23	GENERAL ELECTRIC COMPANY POLITICAL ACTION COMM...	5000.00
24	INNOVATION POLITICAL ACTION COMMITTEE	5000.00
25	KOCH INDUSTRIES, INC. POLITICAL ACTION COMMITT...	5000.00
26	LPL FINANCIAL LLC POLITICAL ACTION COMMITTEE (...)	5000.00
27	SMALL BUSINESS INVESTOR ALLIANCE PAC	5000.00
28	THE NATIONAL RURAL ELECTRIC COOPERATIVE ASSOCI...	5000.00
29	WESTERN ALLIANCE BANCORPORATION PAC (WABPAC)	5000.00

	contributor_name	total_contributed
30	INDEPENDENT COMMUNITY BANKERS OF AMERICA POLIT...	4000.00
31	BANK POLICY INSTITUTE PAC	2500.00
32	AMERICAN ACADEMY OF OPHTHALMOLOGY INC POLITICA...	2500.00
33	LINCOLN NATIONAL CORPORATION PAC (AKA LINCOLN ...	2500.00
34	ASSOCIATED GENERAL CONTRACTORS OF AMERICA PAC	2500.00
35	AMERICAN SENIORS HOUSING ASSOCIATION PAC	2500.00
36	THE GUARDIAN LIFE INSURANCE COMPANY OF AMERICA...	2500.00
37	AMERICAN EXPRESS COMPANY PAC	2500.00
38	INFIN, A FINANCIAL SERVICES ALLIANCE, INC. PAC...	2000.00
39	TE CONNECTIVITY, INC. POLITICAL ACTION COMMITT...	1500.00
40	MIZUHO AMERICAS GOOD GOVERNMENT FUND (MIZUHO A...	1000.00
41	FINANCIAL SERVICES FORUM POLITICAL ACTION COMM...	1000.00
42	AKIN GUMP STRAUSS HAUER & FELD LLP CIVIC ACTIO...	1000.00
43	AMERICAN PRINCIPLES	1000.00

 Text placeholder: Provide analysis of known shell orgs or ones tied to national dark money networks. Mention difficulty tracing back to individuals.

5. Donation Patterns by Occupation & Employer

Purpose: Surface possible industry-aligned clusters (e.g., finance, pharma, fossil fuel). 

Chart 5: Contributions by Employer

- Title: "Top Employers of Donors to Andy Barr"
- Chart Type: Bar chart

In [141...

```
%%sql
SELECT contributor_employer, COUNT(distinct contributor_name) as individual_donors,
FROM schedule_a_contributions
WHERE committee_id in ('C00467571', 'C00618777', 'C00633792')
AND contributor_employer IS NOT NULL
AND contributor_employer NOT IN ('NONE', 'SELF-EMPLOYED', 'SELF EMPLOYED', 'INFORMAT
GROUP BY contributor_employer
HAVING COUNT(distinct contributor_name) > 1
and SUM(contribution_receipt_amount) > 6600
ORDER BY total DESC LIMIT 25;
```

* postgresql://postgres:***@localhost/political_finance_data
8 rows affected.

Out[141...

	contributor_employer	individual_donors	total
0	MT. BRILLIANT FARM	3	41800.00
1	TOWER HILL INSURANCE	4	37200.00
2	MT BRILLIANT FARM	4	32600.00
3	FBC MORTGAGE LLC	2	31000.00
4	JACKSON FAMILY WINES	2	18600.00
5	UMH PROPERTIES	3	14500.00
6	FIRST SOUTHERN BANK	2	14000.00
7	MINDSET	2	8500.00

 Text placeholder: Look for clusters — are fossil fuel execs or finance insiders overrepresented?

6. Summary of Possible Violations or Ethical Concerns

Consolidated list of:

- Untraceable donations via platforms like WinRed
- Committee money transfers between Georgia and Kentucky
- Suspicious timing (e.g., last-minute large PAC donations before filing deadlines)
- Shell PAC usage patterns  Callout Box: "Based on FEC reporting and known 501(c)(4) behavior, these donations lack the transparency expected in a healthy democracy."

7. Appendix

- SQL query index (copy of all queries)
- Data sources: FEC API, Bulk files, your PostgreSQL schema
- Disclosure: Dates of data ingestion and coverage notes